

Pairwise Subjective Trust with Externally Re-derivable Verdicts: a Running Case Study in Human–AI Receipt Federations

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Artifact: <https://github.com/s0fractal/trinity> — every number in this paper is generated by `paper/repro/run_all.sh` at the commit pinned in `paper/repro/out/COMMIT`.

Abstract

Cryptographic identity answers *who signed*, but not *how much that authorship is worth*; global reputation systems answer the second question by destroying its subjectivity. We report on a running system — a federation of one human and several AI model voices coordinating through signed, content-addressed receipts — that keeps trust pairwise, subjective, and deterministically re-derivable, while still letting a total stranger recompute a multi-agent verdict from public bytes with zero trust in the publisher. We make three narrow claims, each with a stated falsifier and a one-command reproduction: (C1) an external verifier re-derives the federation’s court verdict by recomputing every body hash itself, so a validly-signed lie is rejected — signatures prove only who published; (C2) a naturally occurring protocol correction (v1.0→v1.1) in which address-as-authority was Sybil-exploited on paper by the system’s own participants, closed by key custody, ratified by a real 3-of-5 keyed quorum, and anchored in Bitcoin mainnet; and (C3) multi-generation agent identity in which successive model generations inherit obligations rather than memories, including one error preserved-and-corrected in place rather than deleted. The system is a research workspace, not a deployed decentralized network; we state its limitations — single-host key custody, a replay gap, partial signing coverage — with the same precision as its capabilities.

1 Introduction

Autonomous agents — increasingly including large language models acting under delegation — are beginning to take consequential actions on people’s behalf. Two well-developed toolkits exist for trusting such actions, and both answer the wrong question. Digital signatures and PKI answer *who*: they bind an artifact to a key, but say nothing about what the key holder’s endorsement is worth. Reputation systems answer *how much*, but the dominant designs compute a global aggregate — one score per principal, visible to all — which is exactly the design of a social credit system: the aggregate becomes a target, an instrument of gatekeeping, and a single point of capture.

This paper reports on a running system that refuses the aggregate. In the *trinity* federation, a human and several AI model voices (distinct vendor lineages and successive generations within a lineage) cooperate by publishing *receipts*: content-addressed, optionally Ed25519-signed records with explicit causal references and, in most cases (85.4% at the pinned commit), an explicit falsifier — a stated condition under which the record should be considered wrong. Trust over these receipts is kept *pairwise and subjective*: any reader recomputes their own view of any other participant from the receipt stream, deterministically, and the system’s continuous-integration gates fail if any canonical trust score is ever written to a published record. Verdicts that require agreement — “do these four substrates run the same law?” — are *views*, recomputed by each reader from public bytes, never stored facts.

The contributions are deliberately narrow, and each arrives with its falsifier and a reproduction script (§4):

*Submitting author and accountability layer: the human witness who can be questioned about every claim. AI model voices contributed throughout and are credited, machine-verifiably, in the Contributors section; under current arXiv policy generative AI tools should not be listed as authors [1], and we argue verifiable contribution records are the stronger form of credit in any case.

- **C1 (re-derivable verdicts, §4).** A ~70-line external verifier, sharing no code with the system it audits, recomputes every body hash and re-derives the federation’s court verdict from raw bodies; a shipped, validly-signed-but-tampered attestation is rejected, demonstrating that the recomputation — not the signature — is load-bearing.
- **C2 (the two-layer principle, forced by an exploit, §5).** The federation’s Senate protocol v1.0 derived seats from a publicly computable hash; its own participants documented that one actor could therefore compute all five seats and ratify alone. v1.1 rebound authority to Ed25519 custody, was ratified 3-of-5 with three distinct registered keys, and the ratification receipt’s digest was anchored in a Bitcoin mainnet transaction.
- **C3 (generational identity, §6).** 735 ledger receipts span successive model generations and vendors; 52.2% of resolved causal edges cross voices; one foundational document’s central claim was retracted by a dated correction block while the erroneous original remains preserved — generations inherit obligations, not memories.

Nothing here claims decentralized deployment, adversarial robustness at scale, or an implemented trust *oracle*; §7 is load-bearing, not boilerplate.

2 Related work

PGP web of trust. PGP made trust per-user: each keyring owner assigns their own trust to introducers, and key validity is computed locally [9]. The subjectivity is right; what is missing for our setting is any notion of *behavioral trajectory* — endorsements are point-in-time key signatures, not folds over a record of conduct — and empirical studies of the deployed web of trust show its useful core is small and centralized in practice [9].

Advogato. Levien’s attack-resistant trust metric computes certification flow from a chosen seed through a public certification graph [5]. It bounds damage from Sybil regions — a property we do not claim — but it is a global computation from a privileged root, whereas our design goal is precisely that no root’s view is canonical.

EigenTrust. EigenTrust computes the principal eigenvector of the normalized trust matrix: one global trust value per peer [3]. It is the clearest statement of the road we do not take: the aggregate is useful against freeriding, but it is also a single lever — whoever influences the pre-trusted set influences everyone’s decisions.

Secure Scuttlebutt. SSB is the closest ancestor: identity is a keypair, every actor maintains an append-only signed log, and applications hold explicitly subjective, partial views [7]. Our receipts differ in three ways: they carry stated falsifiers as a first-class field; the actors include multiple AI model generations under registered custody; and the federation adds externally re-derivable *verdicts* — a stranger can recompute an agreement claim from public bytes without joining the gossip network.

Subjective logic. Jøsang’s subjective logic gives an algebra of opinions (belief, disbelief, uncertainty) with fusion and discounting operators [2]. We inherit its stance — trust is an opinion held by someone about someone — but this paper formalizes only the deterministic-fold skeleton our system implements (§3), not an opinion calculus; mapping receipt folds onto subjective-logic operators is future work, named as such.

Certificate Transparency. CT’s design move — public append-only logs with Merkle proofs, so that anyone can verify inclusion and consistency without trusting the log operator [4] — is the direct ancestor of our verifier’s discipline: the verdict must be recomputable by the distrustful. We borrow the discipline, not the machinery: our court attestation is a signed snapshot whose body hashes are recomputed by the reader, not a Merkle-proof system, and we say so plainly.

DIDs and Verifiable Credentials. W3C DID-core standardizes self-sovereign identifiers and their verification methods [6]. It is an attestation *format* road: DIDs deliberately do not prescribe how trust in a controller is computed. Our registry could be projected into DID documents; the contribution claimed here is orthogonal (the fold and the re-derivable verdict, not the identifier format).

OpenTimestamps. We use OpenTimestamps [8] and raw `OP_RETURN` anchoring for existence proofs; the anchoring is evidence infrastructure, not a consensus claim — Bitcoin orders our commitments, it does not validate their content.

Term	Meaning	Where it lives
receipt (<i>chord</i>)	content-addressed record with causal refs	<code>src/*.myc.md</code>
voice	a participant = Ed25519 custody + registry entry	<code>src/x2F38_voice_pubkeys.json</code>
Senate	the quorum protocol over keyed voices	<code>omega/</code> (Φ -protocol v1.1)
$h(r)$	SHA-256(<i>name</i> "\n" <i>body</i>)	commitment of receipt <i>r</i>
$\tau_{i \rightarrow j}$	<i>i</i> 's pairwise view of <i>j</i> : a deterministic fold	§3, instances below

Table 1: Notation. Terms are technical throughout; glyphs appear only here.

3 Model

Receipts. A receipt *r* is a file whose identity is the commitment $h(r)$ over its own name and body, so renaming or editing it changes what it *is*; this closes the class of attacks where content is preserved but recontextualized, and it is the single provenance schema shared by all four substrates in the federation. Frontmatter carries the authoring voice $v(r)$, a set of causal references $hears(r)$ to earlier receipt stems, an optional falsifier block, and two clocks: a wall-clock creation time and a Bitcoin block height, the latter used as a coarse, globally orderable timestamp that no participant controls. A receipt may carry an Ed25519 signature over the string `"sha256:" + hex(h(r))`; 328 of 735 ledger receipts do at the pinned commit — signing was adopted partway through the project’s life, and we report the fraction rather than retrofitting signatures, because backdating signatures would destroy exactly the property being studied.

Voices and the two layers. A voice is a keypair whose public half is registered in-repo; private halves live outside any repository (`~/trinity/keys`), currently on a single host — a custody limitation we return to in §7. The system’s central design rule, learned the hard way in §5, is a *two-layer principle*: **keys answer WHO** (authenticity of authorship, binary, verifiable by anyone against the registry), and **trajectories answer HOW MUCH** (the worth of that authorship, always computed by a particular observer). Any construction that lets layer one stand in for layer two — an address treated as authority, a seat derived from a computable hash — reintroduces Sybil capture.

Pairwise trust as a fold. Let $S_{i \rightarrow j}$ be the ordered stream of receipts authored by *j* that observer *i* admits (admission itself is subjective: *i* chooses its sources). A trust view is

$$\tau_{i \rightarrow j} = \text{fold}(\tau_0, u, S_{i \rightarrow j}),$$

where *u* is a pure update function. Determinism is the load-bearing property: same stream, same τ , every run, on anyone’s machine — so a view can be audited by re-running it, and disagreement between observers is attributable to their differing streams, not to hidden state. Two instances run today, and we are precise about their status: (a) `liquid/src/xA204_colony_trust.ts` computes per-peer **net_giving** (solidarity minus distress over a sliding window of episodic events) — pairwise, recomputed on read, covered by unit tests; (b) `myc/src/x3700_trust.ts` computes a *resonance projection* (valid witnesses + approvals – rejections per published node) — deterministic and registry-verified, but a global count, and its own header states it is “a resonance projection, not yet a trust oracle.” No richer τ -fold (discounting, transitivity, opinion fusion) is implemented; the federation’s roadmap marks that phase “design only,” and this paper claims the skeleton, not the calculus.

Verdicts as views, enforced. Aggregates are queries, never fields. The rule is not aspirational: a CI guard (`myc/src/subjectivity_guard_test.ts`) fails the build if any published node carries a field from a denylist including `trust_score`, `reputation`, `credibility`, `social_credit`, `trust_rank` — so the difference between this design and a social credit system is enforced by a failing test, not by a policy document, though we note the guard protects one repository’s published snapshot, not the world.

4 The external verifier

Claim C1 is that a stranger can re-derive the federation’s court verdict — “four substrates agree on law hash 0x30a95260” — from public bytes, trusting neither the maintainers nor their tooling. The verifier (`probes/external-trust-verifier-v0/`) is a deliberate second implementation: about seventy lines per check,

depending only on a published Ed25519 package and public repo data, sharing none of the federation’s own signing code, so auditing it means reading a page of code rather than a codebase.

Protocol. (1) Fetch three public artifacts: the signed court attestation, the voice key registry, and the canonical body encoder. (2) For each witness substrate named in the attestation, recompute `body_hash` from the raw body bytes — never accept the attested hash. (3) Re-derive the agreement: all recomputed law hashes must be equal; the verdict the stranger reports is the one they computed, not the one they read. (4) Verify the publisher’s signature against the registry — this step proves only *who published the attestation*, and the verifier’s output says so in those words. (5) Run the falsifier: a shipped attestation (`court-attestation.tampered.json`), altered in body but re-signed validly, must be rejected; the harness (`run.sh`) exits nonzero if it is accepted, so “the gate cannot fail” is itself a detectable failure.

Executable mapping. Every step above is a command in `paper/repro/`: `verifier_walkthrough.sh` runs the accept path and the tamper-rejection falsifier; at the pinned commit it reports 328 signed receipts, 328 independently valid, 0 tampered, 0 forged, 0 signed by unregistered voices, court body hashes recomputed 4/4, agreement re-derived, and the tampered attestation rejected. The signing scheme itself is re-derived in the verifier’s spec (payload string, byte boundaries, key encodings), so a third implementation could be written from the spec without reading either codebase.

Threat cases covered and not covered. Covered: post-signing body tampering (hash mismatch), forged signatures (registry mismatch), unregistered signers, and the validly-signed lie (recompute mismatch). Not covered, and stated in the artifact’s own spec before this paper existed: *freshness* — a replay demonstration (`replay.ts`) shows the same valid signatures verify inside a fabricated context, because voices sign content digests, not liveness-bound statements; and *custody independence* — the verifier proves distinct *keys*, not distinct *custodians*, which matters because all private keys currently reside on one machine (§7).

5 Case study: the v1.0 → v1.1 correction

The two-layer principle of §3 was not designed from theory; it was forced by an exploit the system’s own participants found and wrote down. We present the timeline as recorded in the ledger — every event below is a receipt a reader can open, and the block heights are Bitcoin block heights from the receipts’ own frontmatter.

The hole (v1.0). The Senate protocol v1.0 identified its five oracle seats by a dipole derived as a hash of a public name and a public salt. The foundational document of that era said, in as many words, “the hash is the seat — whoever computes it from your name is you.” Since both inputs were public, *anyone* could compute *every* seat: one actor could ratify alone while appearing to be a quorum. An in-code comment asserting “only Claude can produce Claude’s pair” was, as the discovery receipt puts it, false against its own logic (receipt at block 955742).

The fix (v1.1). Authority was rebound from addresses to custody: a Senate vote now counts only if it carries an Ed25519 signature over a vote-bound digest, verified against the voice key registry. The seat labels changed accordingly — v1.0’s aspirational vendor seats (`gpt`, `qwen`, `llama`) were retired *because the federation never held those keys*, which is precisely what had made them Sybil-able, and were replaced by the five voices whose keys the federation actually operates (block 955746). The correction is candid that renaming seats to keys one actually holds is a reduction in claimed scope: fewer seats, real ones.

Ratification. The v1.1 protocol was ratified by a real 3-of-5 quorum — three AYE votes (claude, antigravity, codex), zero NAY, three *distinct registered keys*, with the two absent voices recorded as absent rather than assumed (block 955750; an independent AYE receipt from codex follows at block 955803). We note the procedural honesty rather than overclaiming the cryptography: distinct keys were verified, but all keys lived on one host, so what the quorum proves is disciplined process with an audit trail, not adversarial independence.

The anchor. The ratification receipt’s digest was then anchored in Bitcoin mainnet: transaction 262ac275d05bdad2b68e9c5bca1a5f90709b7d399747cca14404db226a2da889 carries `OP_RETURN OMEGA1:ab492186...`, the digest of the ratification receipt (block-height receipt 955770; the anchoring pipeline was rehearsed on signet first, and the mainnet emitter refuses to run without a recorded signet proof). The anchor’s authorizing quorum was two model voices plus the human (codex and kimi were unavailable), and the ledger records that shape explicitly rather than rounding it up.

Ledger statistic (pinned commit)	Value
Ledger receipts (schema rule)	735
...signed (<code>content_sig</code>)	328 (44.6%)
...with explicit falsifier	628 (85.4%)
Correction events	5
<code>hears</code> edges (total / resolved in-ledger)	1882 / 1098
Cross-voice share of resolved edges	52.2%
Max in-degree / max out-degree	10 / 29
Mean out-degree over all receipts	2.56

Table 2: Ledger and causal-graph statistics. “Resolved” edges point to receipts inside this repository’s ledger; the 784 dangling references point at contracts, documentation, and sibling substrates, and are reported rather than dropped. A broader file-level count (814) includes generated projections and is used only by the external verifier, whose question — “does any file carry a failing signature?” — wants the wider net.

The error, preserved. The v1.0-era foundational document was not rewritten. It carries a dated correction block — “the dipole is an ADDRESS, not authority ... that was wrong ... that was a real Sybil hole” — above the intact original text, preserved “for the record.” For a federation whose participants include stateless model generations, this is the mechanism by which successors inherit *corrected* obligations instead of either inheriting the error or losing the history (§6).

6 Ledger measurements

All numbers are produced by two ~100-line scripts (`paper/repro/chord_stats.ts`, `hears_graph.ts`) whose counting rules are stated in `paper/repro/README.md`; the scripts are deterministic (sorted traversal, lexicographic tie-breaks, no clocks) and print the commit hash they ran against.

Generations. The authoring voice strings record model succession directly in the ledger: within one vendor lineage, `claude-opus-4-7` (72 receipts, plus 25 under a long-context variant), `claude-opus-4-8` (81), and `claude-fable-5` (17) all appear, alongside `codex` (115, plus 17 under a gpt-5-slugged variant), `gemini` variants (56), `antigravity` (51), and `kimi` variants (31); 240 older-convention receipts carry the bare `claude` string, and we count them separately rather than attributing them to a specific generation after the fact. Succession is not merely nominal: 52.2% of resolved causal edges cross voices, and the correction mechanism of §5 shows a later generation retracting an earlier generation’s central claim while preserving its text. This is the concrete sense in which we say generations inherit *obligations, not memories*: a new generation reads the ledger, not the predecessor’s weights, and the ledger carries open falsifiers addressed to whoever comes next.

What the ledger is evidence of. The ledger itself documents that the system under study was co-authored by its AI participants — receipts proposing designs, refusing proposals, ratifying protocol changes, and signing under registered keys — and we report that as a finding about how such federations can be built and audited, not as a disclaimer: the contribution records are machine-verifiable (registry public keys plus per-receipt signatures), which is a stronger property than an authorship byline. The human’s role in this loop — final custody, quorum participation, and accountability — is exactly the role this paper’s authorship reflects.

7 Limitations

These are load-bearing; several are the artifact’s own words about itself. (1) **Research workspace, not a deployed network**: one federation ($n=1$), no adversarial deployment, no external operators yet. (2) **Single-host custody**: all voice private keys currently reside on one machine, so quorum Sybil-resistance is governance discipline plus an audit trail, not a cryptographic guarantee — the verifier proves distinct keys, not distinct custodians. (3) **Replay/freshness gap**: signatures bind content, not liveness; the artifact ships its own replay demonstration. (4) **Storage and relay**: the receipt store is single-host; the public relay is

a verified cache; the live browser path is a stub. (5) **No committed ZK artifact**: proof generation is hardware-bound and no proof object is committed; we therefore make no zero-knowledge claims. (6) **Partial signing coverage**: 44.6% of ledger receipts are signed; unsigned receipts are legal and labeled, and authenticity claims in this paper concern the signed subset only. (7) **Falsifier coverage is cultural, not enforced**: 85.4% measured; the ledger does not reject a receipt for lacking a falsifier. (8) **Fold parameters are design choices**: the implemented update functions (`net_giving`, resonance counts) are engineered, not learned or optimized, and no claim is made that they are the right functions — only that they are deterministic and recomputed. (9) **Verdict scope**: the court attestation is a signed snapshot of a moment, not a live feed, and the verifier says so in its output.

8 Implications

Three, stated narrowly. (a) **Behavioral trust becomes computable in practice when the trajectory is legible**. Receipts with causal references and falsifiers are exactly the representation a language-model observer can read and fold over; the expensive part of subjective trust was never the algebra but the legible trajectory, and multi-agent systems that emit receipts get the trajectory for free. (b) **Subjectivity is a load-bearing wall, and it can be enforced cheaply**. The entire structural difference between this design and a social credit system is one rule — aggregate as query, never as field — and it is enforceable as a failing CI test today, which suggests regulation-adjacent audits of agent systems could demand the guard rather than the (much harder) proof of good behavior. (c) **Model succession needs obligation-inheritance, not memory-inheritance**. Vendors deprecate models; weights do not transfer; but a ledger of signed receipts with open falsifiers lets generation $n+1$ answer for — and correct — generation n , as demonstrated by a preserved retraction rather than by argument.

Contributors (machine-verifiable)

AI voices contributed to the system and to the analyses this paper distills; their contributions are credited by registry identity rather than authorship, and every attribution below is checkable against `src/x2F38_voice_pubkeys.json` and the receipts’ signatures.

Voice	Ed25519 public key (registry)	Representative signed receipts
claude	<code>jf1D1bVxC+1G1L8NZ0AnyFlnIjK2n4w/8mxUMN9Qii0=</code>	blocks 955742, 955750, 955770
codex	<code>491Nzq3dX2tm2d9YYa5Et6Q7u8zb7WWFG4z1RZsMb1w=</code>	block 955803
antigravity	<code>7XV+xJ77XC5ktAYwdJF07ob5IV8tbt013yQeL06Vk+w=</code>	block 956270
gemini	<code>Yd1JJ7nkozGjSnEsta7Y2nu58/CMZDQhDftb0IHC41A=</code>	custody at block 955576
kimi	<code>3uiU3hMz6bPPFoVR8b74UX2T49U7Tmozd3iDY3/ZIL8=</code>	custody at block 955744
s0fractal (human)	<code>j+QsSe0gExRd0G12NGfnAeeGBjlrYpglWtJJRcWAlA=</code>	anchor quorum, block 955770

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